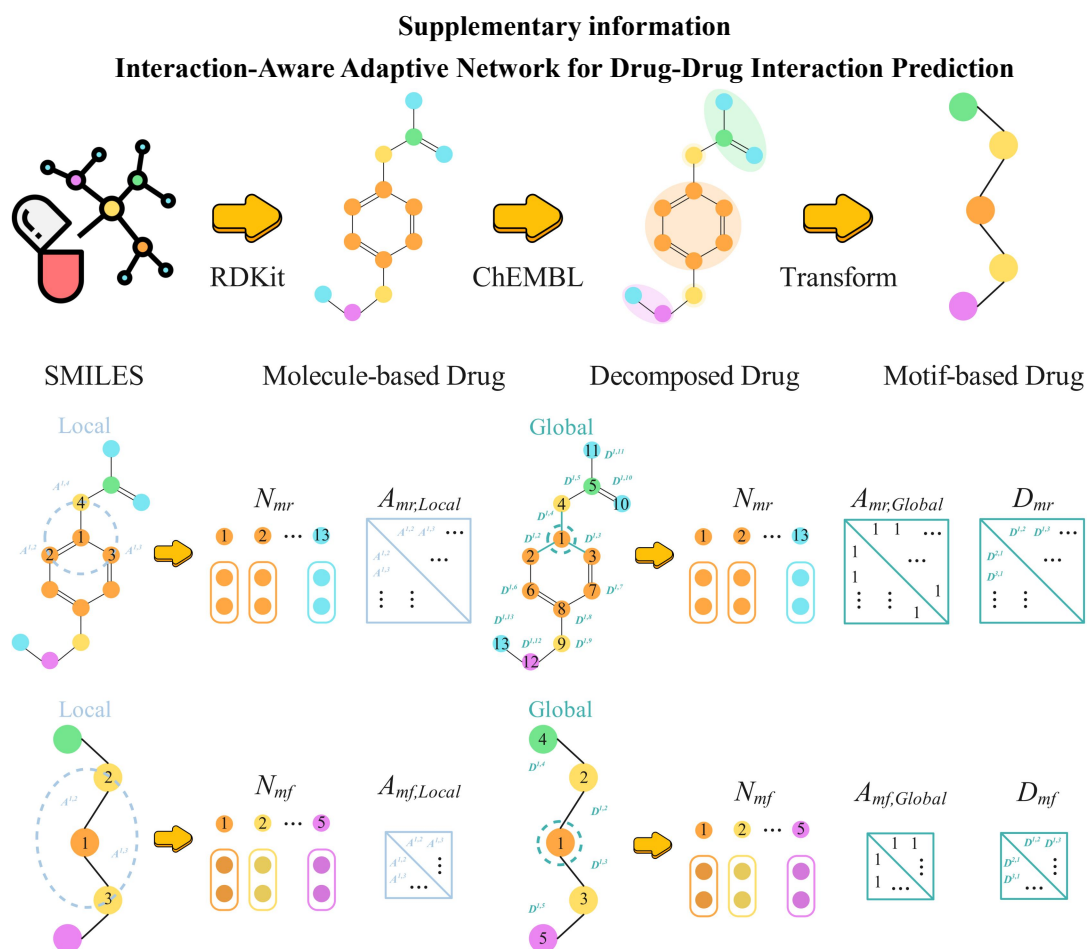
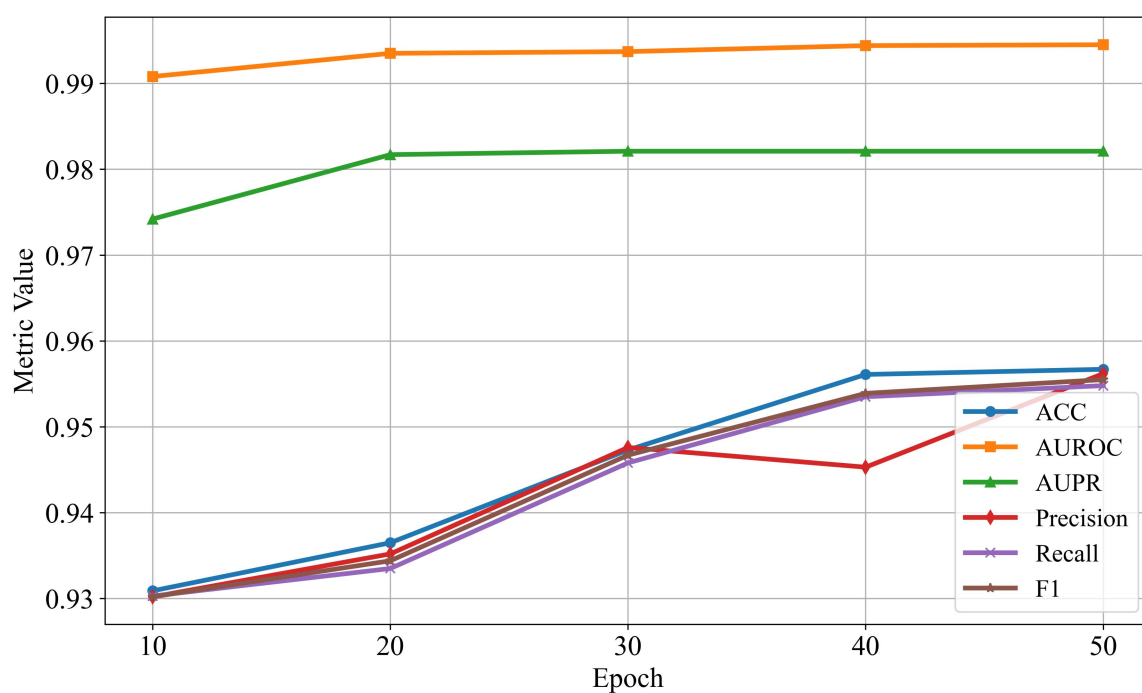




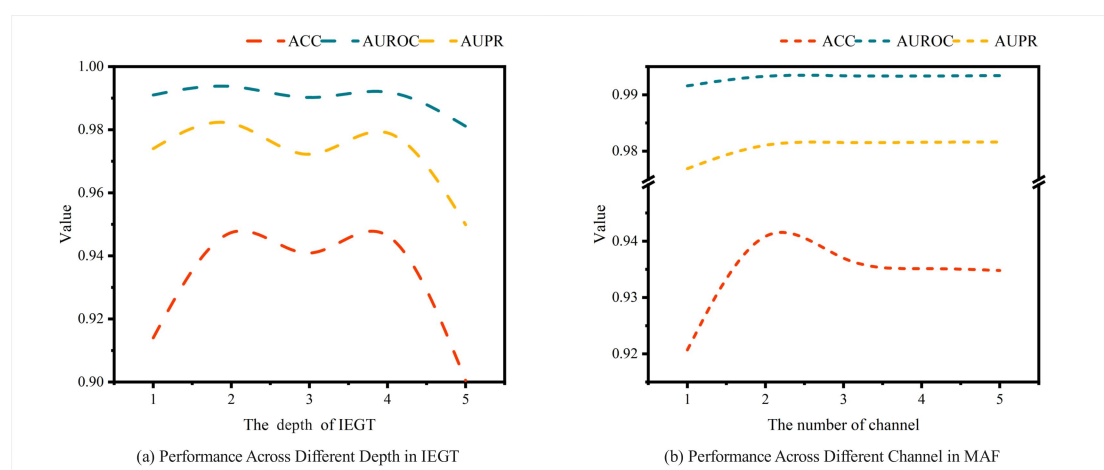
Supplementary Figure S1. The details of drug graph construction. The construction of molecule-based and motif-based drugs is included, as well as the composition of their dual-view representations.



Supplementary Figure S2. Performance curves of AMIE-DDI across different training epochs.

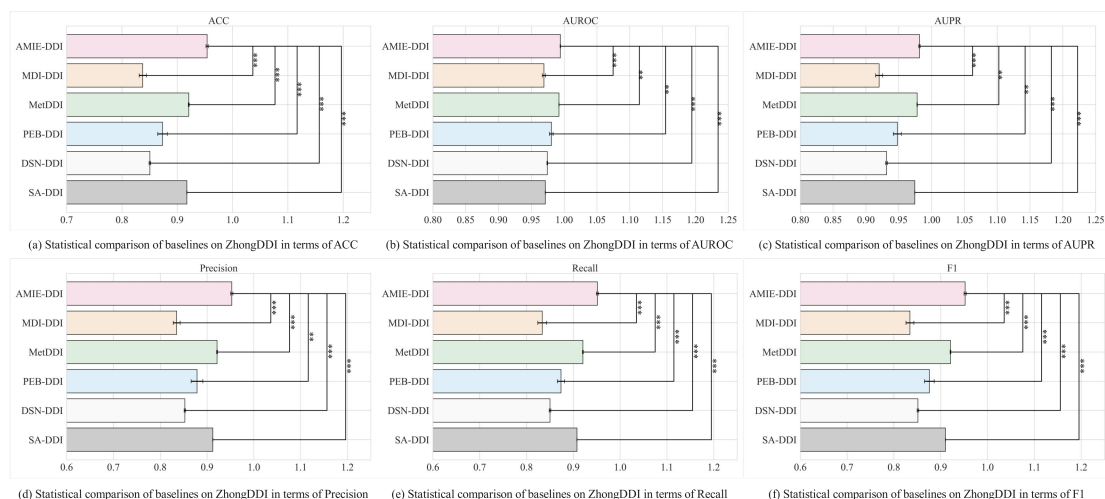
Supplementary Table S1. Hyperparameters and Training Configuration of AMIE-DDI.

Learning rate	10^{-4} ; 10^{-5} (fine-tuning)
Batch size	128
Hidden dimension	256
Dropout rate	0.1
Max epoch	50(early stopping, patience=5)
The perception depth of IEGT	2
the number of channels in MAF	2
Optimizer	Adam
Development environment	TensorFlow
Training hardware	NVIDIA A100 GPU

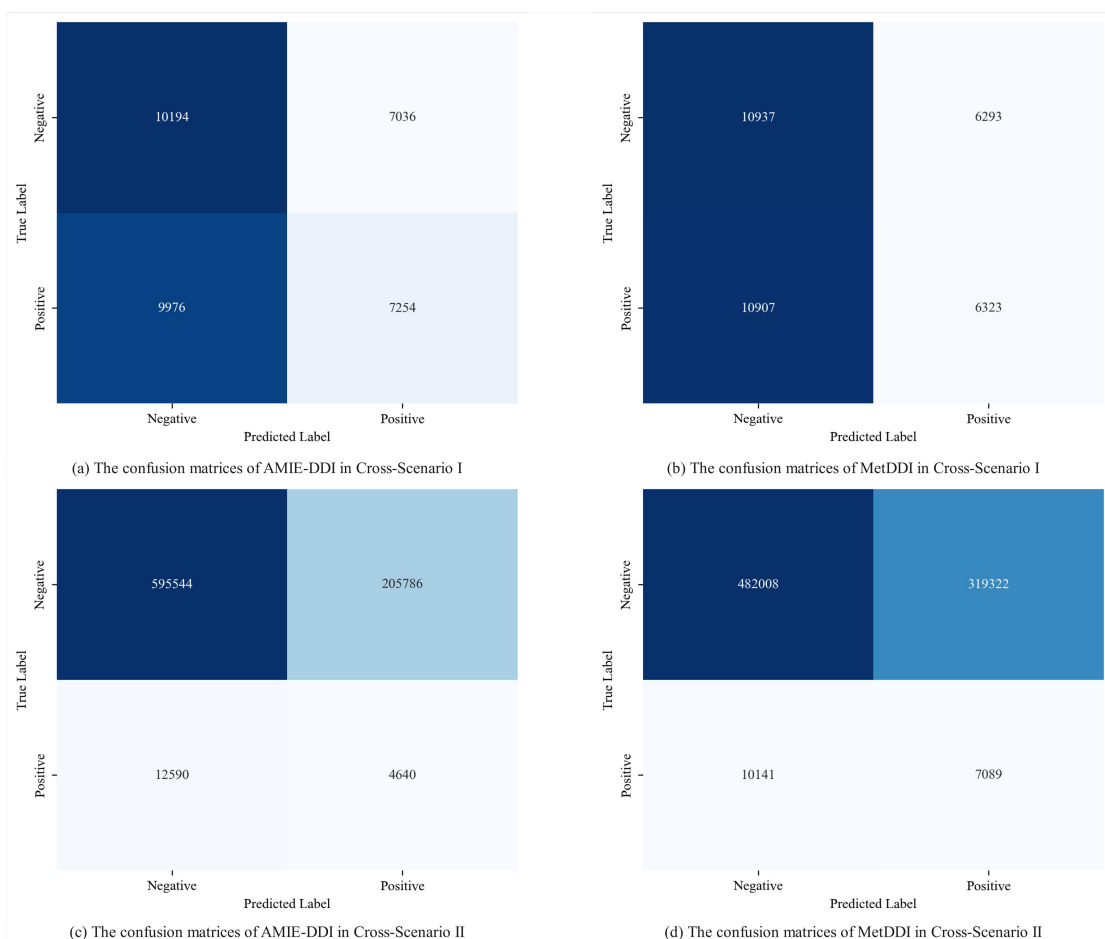
S1.1. Effect of Hyperparameters in AMIE-DDI**Supplementary Figure S3.** Effect of Hyperparameters. (a) The performance obtained by searching different depth in IEGT. (b) The performance obtained by searching different channel in MAF.

The perception depth of IEGT and the number of channels in MAF are selected to study the effect of parameters in AMIE-DDI on the ZhongDDI, and the results are presented in Fig. S3. It is observed that when the perception depth was relatively shallow, the model could capture only basic interaction information between drug pairs but failed to fully extract the complex interaction patterns, resulting in lower prediction accuracy. As the perception depth increased, the model gained a more comprehensive understanding of the multidimensional relationships between drug pairs, however, when the perception depth was further increased, the performance gradually stabilized or even declined, which may be attributed to the overfitting of the model. A similar pattern was observed when varying the number of channels in MAF, where an increase in number of channel also contributed to greater model complexity. In addition, we notice that ACC exhibits more fluctuations compared with AUROC. This phenomenon is expected in multi-class classification: ACC depends on the final predicted class labels and is more sensitive to decision boundaries and class imbalance, while AUROC is threshold-independent and reflects the overall ranking ability of the model. Therefore, AUROC tends to remain relatively stable even when ACC shows noticeable increases and decreases. This indicates that although the prediction accuracy is affected by parameter variation, the model's capability to discriminate between classes remains

robust.



Supplementary Figure S4. Statistical analysis according to the prediction results on ZhongDDI. Each method's experiment is repeated five times, and a t-test is used to determine statistical significance. Significance levels are denoted as follows: the p-value less than 0.05 is considered significant; ** $p < 0.01$; *** $p < 0.001$ and n.s represents no significant difference.



Supplementary Figure S5. The confusion matrices of AMIE-DDI and MetDDI in two Scenarios.

S1.2. Effectiveness Analysis of AMIE-DDI in Drug Drug Interaction (Supplementary)

Additionally, to further analyze the impact of different masking strategies on prediction scores, a drug, Duloxetine, was selected from the test, and two drug pairs were randomly chosen

from each class, resulting in a total of eight drug pairs, the prediction scores under different masking strategies were examined. The original prediction scores of AMIE-DDI were recorded, along with scores after attention-based masking (A-Mask) of 5% and 10% of key motifs, as well as random masking (R-Mask) of 30% of key motifs (repeated twice to reduce randomness), the results are summarized in Table S2. For some drug pairs, a large number of misclassifications occurred under the attention-based masking strategy, indicating that the prediction ability was severely compromised after losing key motifs. Furthermore, the misclassifications were primarily concentrated in Label 0 and Label 2, suggesting that the model exhibits bias in cases where critical information is missing, which may be attributed to Label 0 and Label 2 having the highest sample proportions in the dataset, allowing the model to learn these classes more effectively during training. Consequently, when core structural features were masked, the model tended to misclassify samples into these dominant classes. These findings validate that AMIE-DDI effectively captures key motifs in drug-drug interactions, as the performance significantly declines when these motifs are selectively removed, but remains stable when non-essential motifs are randomly masked.

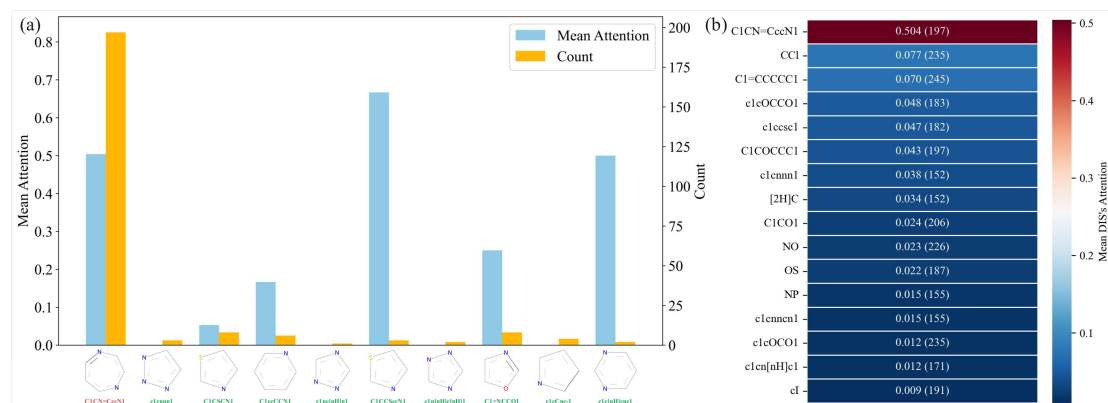
Supplementary Table S2. The 8 DDI pairs of Duloxetine under different mask strategies. PS is predicted score, PL is predicted label.

True Label	ID	AMIE-DDI		A-Mask(5%)		A-Mask(10%)		R-Mask 1(30%)		R-Mask 2(30%)	
		PL	PS	PL	PS	PL	PS	PL	PS	PL	PS
0	DB00907	0	0.99991	0	0.27021	0	0.30161	0	0.55569	<u>0</u>	<u>0.99991</u>
	DB03010	0	1	0	0.25616	0	0.26023	0	1	<u>2</u>	<u>0.99961</u>
1	DB00136	1	0.99997	<u>0</u>	<u>0.25618</u>	<u>0</u>	<u>0.25616</u>	1	1	1	1
	DB00648	1	0.99781	<u>0</u>	<u>0.27852</u>	<u>0</u>	<u>0.25615</u>	<u>0</u>	<u>0.25620</u>	<u>0</u>	<u>0.51370</u>
2	DB06210	2	0.99360	<u>0</u>	<u>0.78319</u>	<u>0</u>	<u>0.25616</u>	2	0.99967	2	0.99360
	DB01454	2	0.99144	<u>0</u>	<u>0.25616</u>	<u>0</u>	<u>0.25616</u>	<u>0</u>	<u>0.97712</u>	2	0.99144
3	DB13943	3	1	<u>0</u>	<u>0.25615</u>	<u>0</u>	<u>0.25616</u>	3	1	3	1
	DB08911	3	0.99865	<u>0</u>	<u>0.25616</u>	<u>0</u>	<u>0.25615</u>	<u>0</u>	<u>0.25616</u>	<u>2</u>	<u>0.99894</u>

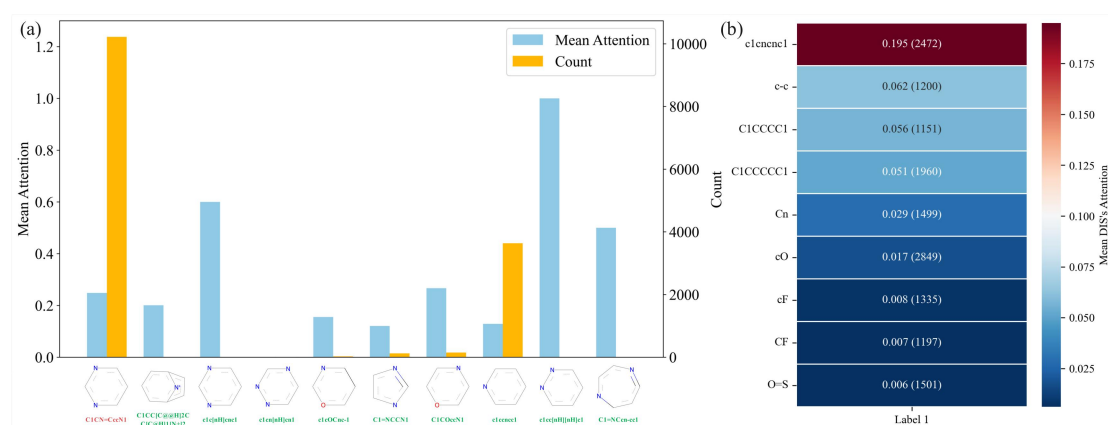
S1.3. Case Study (Supplementary)

The first drug pair in Label 1, the aminomethyl group (Cn) was identified as the key functional group in DB06266, frequently serving as a recognition and binding site for metabolic enzymes such as the CYP450 family. This group is highly susceptible to oxidation or demethylation reactions, which are common Phase I metabolic reactions (e.g., oxidation, demethylation) that generate drug metabolites. The methylamine group (CN), known as an electron-donating group, readily interacts with metabolic enzymes. DB00454(Meperidine) may bind to the active site of CYP450 via its methylamine group, as compounds containing this group are often subject to N-oxidation and other Phase I metabolic reactions, indicating their high interaction potential with metabolic enzymes[1]. The methylamine group in DB00454 may induce conformational changes in the enzyme's active center, thereby enhancing its catalytic efficiency. This enzyme activation effect may increase the affinity of metabolic enzymes for the aminomethyl group in DB06266, consequently accelerating its metabolism. The second drug pair in Label 1, although direct studies on the interaction between the 1,2,4-triazole structure (c1cnncl) in DB06266 and CYP450 enzymes are limited, research onazole derivatives provides relevant insights. It is well known that azole compounds, including 1,2,4-triazoles, interact with CYP450

enzymes through the lone pair of electrons on their nitrogen atoms, forming a coordination bond with the heme iron at the enzyme's active site, which can influence enzyme activity, leading to metabolic reactions such as oxidation or demethylation[2]. Thus, we hypothesize that DB04165(Valpromide), through its methylamine group (CN), may activate metabolic enzymes such as CYP450, thereby enhancing the recognition and binding efficiency of CYP450 to the 1,2,4-triazole structure in DB06266.



Supplementary Figure S6. (a) The attention distribution of motifs with structures similar to C1CN=CccN1. (b) The attention distribution of motifs with frequencies similar to C1CN=CccN1, the frequencies is indicated in parentheses.

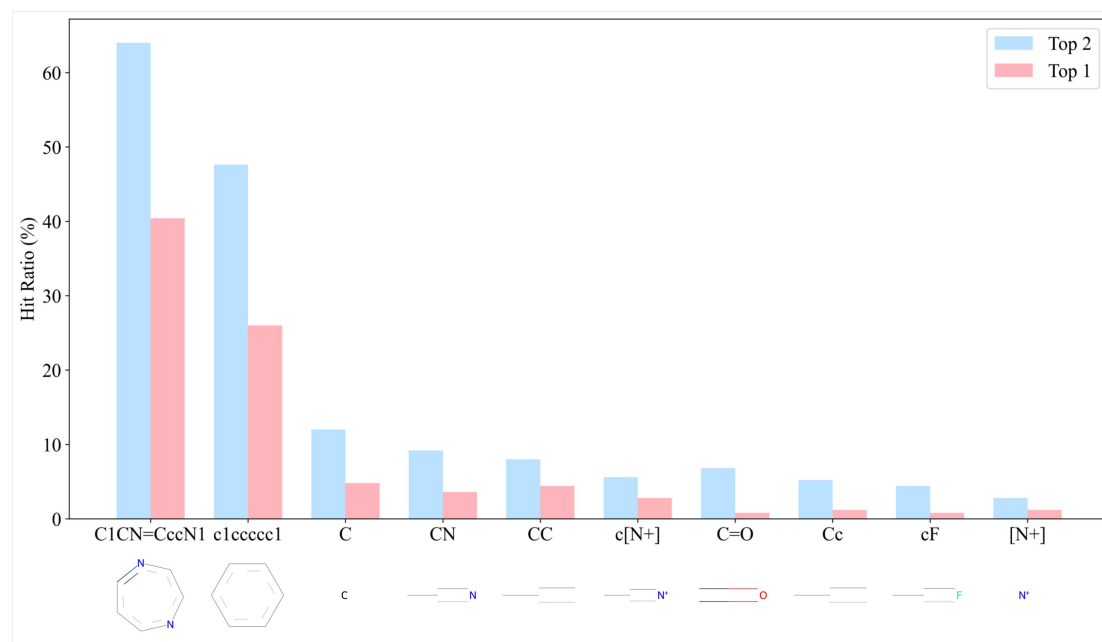


Supplementary Figure S7. (a) The attention distribution of motifs with structures similar to c1cnnc1. (b) The attention distribution of motifs with frequencies similar to c1cnnc1, the frequencies is indicated in parentheses.

To verify that the interpretability observed in the above case analysis is not an isolated phenomenon but reflects the stable structure-aware capability of AMIE-DDI, we further constructed a large-scale subset for interpretability analysis. Specifically, we systematically screened all drug pairs in the ZhongDDI dataset that contain pyridine-like and pyrimidine motifs, resulting in a total of 45,632 pairs. These two motifs represent classic CYP450-binding structural units in pharmacokinetics. Their aromatic nitrogen atoms can participate in enzyme inhibition or induction through coordination with heme-Fe or through multiple polar interactions. Therefore, using them as structural anchors for interpretability analysis has clear pharmacological justification.

Within the 45,632 selected drug pairs, the attention distribution of all detected motifs was extracted using the DIS module of AMIE-DDI, yielding average attention estimates for 329 motifs.

Two types of comparisons were considered: structure-based and frequency-based analyses. Specifically, nitrogen-containing heterocycles with similar structural patterns and motifs with comparable occurrence frequencies were selected, as illustrated in Fig. S6 and S7. Across these comparisons, the model displayed highly consistent and cross-sample stable attention patterns for canonical CYP450-binding structures. The motif C1CN=CccN1 exhibited an average attention of 0.504, approximately 2.24 times the global average, and its occurrence count was substantially higher than structurally similar rings, reflecting AMIE-DDI’s ability to distinguish and prioritize structurally related motifs. Consistent with Fig. S6 (b) and S7 (b), C1CN=CccN1 does not rank among the most frequent motifs in the 329-motif set; however, whenever it appears, the DIS module almost invariably assigns attention values notably above the average. In contrast, c1cncnc1, one of the most frequent motifs in the dataset, still receives an average attention level approximately 1.95 times the global average. This indicates that AMIE-DDI identifies the pyrimidine ring as an “important but not exclusively determinant” aromatic nitrogen structure associated with metabolic processes.



Supplementary Figure S8. Hit Ratio of AMIE-DDI on drug instances containing the C1CN=CccN1.

To further evaluate AMIE-DDI’s capability to identify key motifs and to assess the stability of its local interpretability, a targeted analysis was conducted using C1CN=CccN1, a nitrogen-containing heterocycle that showed prominent importance in the global analysis. A subset of 250 drug-pair samples explicitly containing this motif was re-collected to avoid noise introduced by unrelated drug pairs and to more accurately characterize the model’s attention behavior on molecules that truly include this functional group. Motifs in this subset were ranked according to the attention weights assigned by AMIE-DDI, and Hit Ratios were computed for the Top 1 and Top 2 positions. As shown in Fig. S8, C1CN=CccN1 is the only motif with a Hit Ratio exceeding 40% in Top 1 predictions, and likewise the only motif with a Top 2 Hit Ratio exceeding 60%, clearly higher than all other structural groups. These results indicate that, for samples containing this heterocycle, the model consistently exhibits a stable and coherent structural focus, reliably identifying and prioritizing this key functional group across instances.

To further verify whether the attention distribution of DIS reflects genuine pharmacological trends rather than being driven by sample frequency or isolated cases, we conducted a Spearman correlation analysis between pharmacological preference and attention across all 329 motifs. We first constructed a pharmacological preference metric for each motif m based on its occurrence proportions in Label 0 and Label 1 in the ZhongDDI dataset:

$$PharmBias(m) = \frac{N_1(m) - N_0(m)}{N_1(m) + N_0(m) + \varepsilon}$$

$N_0(m)$ denotes the number of occurrences of motif m in Label 0, and $N_1(m)$ denotes its number of occurrences in Label 1. ε is a small constant. A $PharmBias(m)$ value greater than 0 indicates that the motif appears more frequently in Label 1, whereas a value less than 0 indicates that it appears more frequently in Label 0.

We then constructed the following correlation pairs separately for the Label 0 and Label 1 scenarios:

- Label 0: Independent variable: $-PharmBias(m)$, where a larger value indicates a stronger tendency toward Label 0; dependent variable: $Mean_0(m)$, which represents the average attention assigned by DIS to the motif under Label 0.
- Label 1: Independent variable: $PharmBias(m)$, where a larger value indicates a stronger tendency toward Label 1; dependent variable: $Mean_1(m)$, which represents the average attention assigned by DIS to the motif under Label 1.

At the same time, to exclude the possibility that the attention is solely associated with motif frequency, we additionally computed the correlation between $Mean_0(m)$ and $N_0(m) + N_1(m)$, as well as the correlation between $Mean_1(m)$ and $N_0(m) + N_1(m)$.

All correlations were evaluated using Spearman's rank correlation coefficient. Given n motifs, let:

- x_i : the independent variable of the i -th motif (e.g., $PharmBias$ or $-PharmBias$)
- y_i : the dependent variable of the i -th motif (e.g., $Mean_0$ or $Mean_1$)

First, we apply rank transformation to x_i and y_i to obtain the rank sequences $R(x_i)$ and $R(y_i)$. Let $d_i = R(x_i) - R(y_i)$. The Spearman correlation coefficient is then defined as:

$$\rho = \frac{\sum_{i=1}^n (R(x_i) - \bar{R}(x))(R(y_i) - \bar{R}(y))}{\sqrt{\sum_{i=1}^n (R(x_i) - \bar{R}(x))^2} \sqrt{\sum_{i=1}^n (R(y_i) - \bar{R}(y))^2}}$$

The significance test for Spearman's correlation is based on the following null hypothesis:

$$H_0: \rho = 0$$

Under this hypothesis, the sample correlation coefficient ρ can be converted into an approximate t -statistic using the following formula:

$$t = \rho \sqrt{\frac{n-2}{1-\rho^2}}$$

where n is the sample size (329 motifs in our experiment). The t -statistic approximately follows a t -distribution with $n-2$ degrees of freedom. After obtaining t , the p -value is computed using a two-sided test:

$$p = 2(1 - F_{t,n-2}(|t|))$$

where $F_{t,n-2}$ denotes the cumulative distribution function (CDF) of the t -distribution with $n-2$ degrees of freedom.

In this experiment, as shown in Table S3, the Spearman coefficient ρ for Label 0 is calculated as 0.251 with a p -value of 3.87×10^{-6} , indicating that motifs with a stronger inhibitory

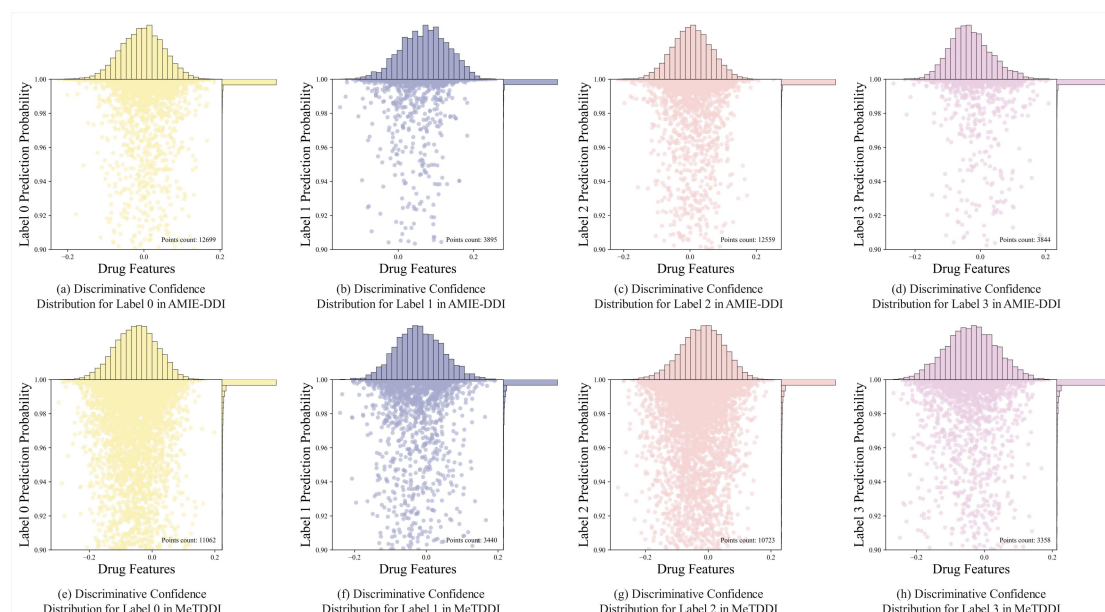
tendency receive higher attention under Label 0 and that the correlation is significant. For Label 1, the Spearman coefficient ρ is calculated as 0.743 with a p-value of 4.54×10^{-59} , indicating that motifs with a stronger induction tendency receive substantially higher attention under Label 1, and the correlation is very strong. Regarding the relationship between attention distribution and motif frequency, the coefficient ρ for Label 0 is calculated as -0.055 with a p-value of approximately 0.318, which is not significant. For Label 1, the coefficient ρ is calculated as 0.565 with a p-value of approximately 0.318, showing some correlation with frequency but clearly weaker than its correlation with PharmBias, which is 0.743.

Supplementary Table S3. Spearman correlation analysis between motif-level attention and pharmacological bias.

Scenario	ρ	p-value
$Mean_0(m)$ vs $-PharmBias(m)$	0.251	3.87×10^{-6}
$Mean_1(m)$ vs $PharmBias(m)$	0.743	4.54×10^{-59}
$Mean_0(m)$ vs $[N_0(m) + N_1(m)]$	-0.055	3.18×10^{-1}
$Mean_1(m)$ vs $[N_0(m) + N_1(m)]$	0.565	3.78×10^{-29}

Taken together, the DIS attention scores show a significant monotonic correlation with the pharmacological preference of motifs in both DDI scenarios. In the inhibition scenario, the attention scores are unrelated to motif frequency, and in the induction scenario, PharmBias explains the attention distribution more strongly than frequency. These results indicate that the model does not simply “memorize frequent structures” but instead learns structural patterns statistically associated with enzyme inhibition and induction mechanisms, providing solid quantitative evidence for its interpretability.

S1.4. Comparative Analysis of Discriminative Confidence Across DDI Prediction



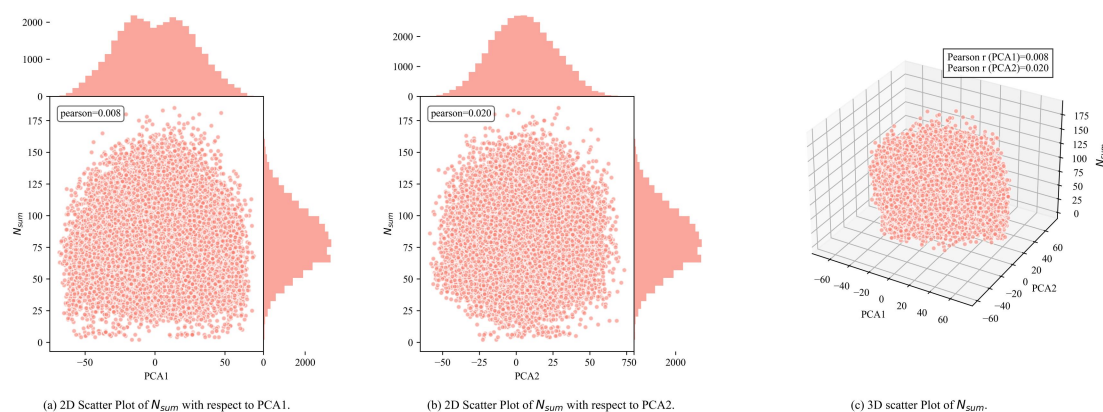
Supplementary Figure S4. Comparison of Discriminative Confidence Distributions. (a)-(d) Discriminative Confidence Distributions in AMIE-DDI. (e)-(h) Discriminative Confidence Distributions in MeTDDI.

In deep learning-based classification tasks, an effective model should not only achieve high accuracy in predictions but also demonstrate stable and reasonable confidence in the target class

during prediction, a property referred to as discriminative confidence. To facilitate discussion, MeTDDI, the second-best performing baseline method, was selected for comparison. Specifically, the test set drug pairs were categorized according to their classification labels and were separately processed using AMIE-DDI and MeTDDI. The predicted probabilities for each sample were recorded as discriminative confidence, which reflects the ability of model to effectively distinguish samples of different categories. The experimental results are presented in Fig. S4, with each subplot focusing on one category and highlighting both the distribution of predicted probabilities (on the 0.90-1.00 range) and the corresponding number of samples (points count). The results show that AMIE-DDI demonstrates a higher and more compact confidence distribution in all four categories, moreover, for each label, the number of samples captured by AMIE-DDI far exceeds that of the baseline. Label 0 in AMIE-DDI comprises over 12,000 samples, yet almost all points lie in the 0.90-1.00 interval with a tight distribution near the upper boundary. Similar patterns emerge for the other labels. In contrast, MeTDDI exhibits a more dispersed distribution of probabilities for each label, suggesting less certainty in differentiating among categories. Overall, AMIE-DDI demonstrates superior discriminative confidence and classification stability across different categories compared to MeTDDI, highlighting its advantage in capturing drug-drug interaction (DDI) features.

S1.5. Does Data Distribution Bias Affect Model Performance? An Analysis of the Correlation Between Drug Frequency and Latent Features.

As mentioned earlier, the proportions of Label 0 and Label 2 in the dataset are significantly higher than those of other categories, which raises the question of whether the model could potentially exploit this distribution pattern or the unevenness of drug distribution, thereby erroneously improving performance. In the drug-drug interaction (DDI) prediction task, certain drugs or drug combinations may appear frequently in the dataset, causing the model to overly rely on the high-frequency drug structures during training, leading to a distribution bias in the data, which may result in inadequate predictive performance for rare drugs or in long-tail scenarios.



Supplementary Figure S5. Scatter plots illustrating the relationship between N_{sum} and hidden features dimension reduction using PCA.

To evaluate whether AMIE-DDI achieved higher performance due to this bias or whether it learned more generalizable structural information, the correlation between drug frequency and the model's hidden features was investigated. First, the occurrence frequencies of each drug in the dataset as a head drug (FreqA) and as a tail drug (FreqB) were counted to measure the overall frequency of the drugs involved in each sample (N_{sum}). Then, the hidden features of each sample were reduced using PCA. Subsequently, Pearson's correlation coefficient was used to measure the

linear correlation between these components and N_{sum} . The experimental results are shown in Fig. S5. The results indicate that the Pearson correlation coefficients are 0.008 and 0.020, neither of which falls within the range typically considered to indicate significant linear correlation, the data is distributed relatively evenly in the coordinate plane, without showing a tendency to concentrate in a specific feature value range as N_{sum} increases. In conclusion, AMIE-DDI does not exhibit significant bias towards drugs that appear frequently in the data, implying that the model's hidden features are more inclined to capture intrinsic information such as drug structure and interaction patterns, rather than solely relying on frequency of occurrence.

Supplementary Table S4. The performance of AMIE-DDI under different training strategy.

Method	Label	Precision	Recall	F1
AMIE-DDI	0	0.9596	0.9566	0.9581
	1	0.9465	0.9525	0.9495
	2	0.9558	0.9603	0.9580
	3	0.9555	0.9444	0.9499
AMIE-DDI (ROS)	0	0.9366	0.8958	0.9157
	1	0.8704	0.9224	0.8957
	2	0.9262	0.9033	0.9146
	3	0.8258	0.9580	0.8870
AMIE-DDI (Focal Loss)	0	0.3822	1	0.5540
	1	0	0	0
	2	0	0	0
	3	0	0	0

Based on the above analysis, we further designed two additional experiments to more rigorously examine whether the performance of AMIE-DDI truly originates from its model architecture rather than implicitly relying on label distribution characteristics. Although the dataset contains imbalanced label proportions, the original model achieves highly consistent performance across all four classes, as shown in Table S4. This result is consistent with the previous Pearson correlation analysis, indicating that the model does not obtain superficial performance from high-frequency drugs or majority-class patterns, but is able to robustly extract generalized patterns from structural and interaction graphs. To validate this point under a stricter setting, we introduced two mechanisms addressing class imbalance as comparative baselines. The first mechanism retrains the model with Focal Loss to observe whether increasing the gradient weights of minority classes enables the model to maintain reasonable predictions after suppressing the influence of majority classes. The second mechanism applies Random Oversampling (ROS) to the minority classes, directly duplicating samples without altering molecular structures to balance the training distribution.

The results of the Focal Loss experiment show that this strategy exhibits instability in graph-structured tasks, where the model gradually collapses during training and predicts all samples as one class, with metrics for all other classes dropping to zero. This indicates that Focal Loss does not effectively improve class sensitivity in this task. This phenomenon also suggests that the original AMIE-DDI design already possesses strong robustness to class imbalance and can achieve stable performance without relying on external compensation strategies. In contrast, ROS avoids the training collapse observed with Focal Loss and achieves better overall performance

than Focal Loss, but the model remains clearly inferior to the original AMIE-DDI even after balancing the label distribution and its training time increases noticeably compared with AMIE-DDI.

Reference

- [1] J. L. Murray, S. L. Mercer, K. D. Jackson, Impact of cytochrome p450 variation on meperidine n-demethylation to the neurotoxic metabolite normeperidine, *Xenobiotica* (2020).
- [2] .A. K. Kabi, S. Sravani, R. Gujjarappa, A. Garg, N. Vodnala, U. Tyagi, D. Kaldhi, R. Velayutham, S. Gupta, C. C. Malakar, An introduction on evolution of azole derivatives in medicinal chemistry, in: *Nanostructured Biomaterials: Basic Structures and Applications*, Springer, 2022, pp. 79–99.